Date

积分

1. (Jensen. Riemann和) 设 $f\in R[a,b]$, $\forall x\in[a,b]$, $m\le f(x)\le M$, if $\varphi\in C[m,M]$ 下凸,

证: $\varphi\left(\frac{1}{b-a}\int_a^b f\right)\le \frac{1}{b-a}\int_a^b(\varphi\circ f)$

证: $\because \varphi\in C$ 且 $f\in R$ $\therefore \varphi\circ f\in R$

$$\varphi\left(\frac{1}{b-a}\int_a^b f\right)=\varphi\left(\frac{1}{b-a}\lim_{\lambda(P)\to 0}\sum_{i=1}^{n} f(\xi_i)\Delta x_i\right)=\lim_{\lambda(P)\to 0}\varphi\left(\sum_{i=1}^{n}\frac{\Delta x_i}{b-a}f(\xi_i)\right)\le \lim_{\lambda(P)\to 0}\sum_{i=1}^{n}\frac{\Delta x_i}{b-a}f(\xi_i)$$

相对系数, 且 $\Sigma=1$

$$=\frac{1}{b-a}\lim_{\lambda(P)\to 0}\sum_{i=1}^{n} f(\xi_i)\Delta x_i=\frac{1}{b-a}\int_a^b\varphi(f(x))\,dx$$

② 加强: 设 $p(x)\ge 0$, $\int_a^b p>0$, 则有 $\varphi\left(\frac{1}{\int_a^b p}\cdot\int_a^b(f\cdot p)\right)\le\frac{1}{\int_a^b p}\int_a^b\varphi(f(x))\cdot p(x)\,dx$

证: $\varphi\left(\frac{1}{\int_a^b p}\cdot\int_a^b f\cdot p\right)=\varphi\left(\frac{1}{\int_a^b p}\lim_{\lambda(P)\to 0}\Sigma f(\xi_i)\,p(\xi_i)\Delta x_i\right)$

$$=\lim_{\lambda(P)\to 0}\varphi\left(\sum_{i=1}^{n}\frac{p(\xi_i)\Delta x_i}{\int_a^b p}\cdot f(\xi_i)\right)\le\lim_{\lambda(P)\to 0}\sum_{i=1}^{n}\frac{p(\xi_i)\Delta x_i}{\int_a^b p}\varphi(f(\xi_i))$$

系数

$$=\frac{1}{\int_a^b p}\cdot\lim_{\lambda(P)\to 0}\sum_{i=1}^{n}\varphi(f(\xi_i))\cdot p(\xi_i)\cdot\Delta x_i=\frac{1}{\int_a^b p}\cdot\int_a^b\varphi(f(x))p(x)\,dx$$

③ esp. if $\int_a^b f(x)\,dx=1$. 有 $\varphi\left(\int_a^b f(x)p(x)dx\right)\le\int_a^b\varphi(f(x))\,p(x)\,dx$

④ 概率语言中, 有: $\varphi(E[f(x)])\le E[\varphi\circ f(x)]$

2. (凸) (1) 设 f $[a,b]$ 下凸 $\forall x_1<x_2\in(a,b)$, 证:

f不一定可导! ∴用定义式: $f(\lambda x+(1-\lambda)x_1)\le\lambda f(x)+(1-\lambda)f(x_1)$

$$f\left(\frac{x_1+x_2}{2}\right)\le\frac{1}{x_2-x_1}\int_{x_1}^{x_2}f(x)\,dx\le\frac{f(x_1)+f(x_2)}{2}$$

证: ① $\int_{x_1}^{x_2}f(\lambda x_1+(1-\lambda)x_2)\,d\lambda\le\int_{x_1}^{x_2}\lambda f(x_1)+(1-\lambda)f(x_2)\,d\lambda$

$$\Rightarrow\frac{1}{x_2-x_1}\int_{x_1}^{x_2}f\le\frac{f(x_1)+f(x_2)}{2}$$

② $\int_{x_1}^{x_2}f(x)\,dx=\int_{x_1}^{x_2}\frac{f(x)+f(x_1+x_2-x)}{2}\,dx\ge\int_{x_1}^{x_2}f\left(\frac{x_1+x_2}{2}\right)dx=(x_2-x_1)f\left(\frac{x_1+x_2}{2}\right)$

$$\Rightarrow f\left(\frac{x_1+x_2}{2}\right)\le\frac{1}{x_2-x_1}\int_{x_1}^{x_2}f(x)\,dx\le\frac{f(x_1)+f(x_2)}{2}$$

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(没说凸)

(2) 设 $f\in C^2[a,b]$, 证: $\exists\xi\in[a,b]$, s.t. $\int_a^b f(x)dx=(b-a)f(\frac{a+b}{2})+\frac{1}{24}(b-a)^3f''(\xi)$

$\exists\eta\in[a,b]$, s.t. $\int_a^b f(x)dx=(b-a)\frac{f(a)+f(b)}{2}-\frac{1}{12}(b-a)^3f''(\eta)$

证: $\because f''\in C[a,b]$ $\therefore$ 由最值, f'' 在 $[a,b]$ 上 $\exists$ $\min=M_1$, $\max=M_2$

令 $h_i(x)=f(x)-\frac{M_i}{2}(x-\frac{a+b}{2})^2$ $h_1''\geq 0$, $h_2''\leq 0$

$$\Rightarrow\begin{cases} h_1(\frac{a+b}{2})\leq\frac{1}{b-a}\int_a^b h_1\leq\frac{h_1(a)+h_2(b)}{2} \\ \frac{h_2(a)+h_2(b)}{2}\leq\frac{1}{b-a}\int_a^b h_2\leq h_2(\frac{a+b}{2}) \end{cases}\Rightarrow$$

$$\Rightarrow\frac{\int_a^b f}{b-a}\in\left[f(\frac{a+b}{2})+\frac{(b-a)^2}{24}M_1,\ f(\frac{a+b}{2})+\frac{(b-a)^2}{24}M_2\right]$$

由介值 $\exists\xi$,

$$\in\left[\frac{f(a)+f(b)}{2}-\frac{(b-a)^2}{12}M_2,\ \frac{f(a)+f(b)}{2}-\frac{(b-a)^2}{12}M_1\right]$$

$\exists\eta$.

3.(1) 第一积分中值 Thm (弱): 设 $f\in C[a,b]$. 则 $\exists\xi\in[a,b]$. s.t. $\frac{\int_a^b f(x)dx}{b-a}=f(\xi)$

加权: 设 $f\in C[a,b]$, $g\in R[a,b]$, 且 $g\geq 0$, $\int_a^b g(x)dx>0$, 则 $\exists\xi\in(a,b)$, s.t. $\frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx}=f(\xi)$

g不变号且不恒=0!

证: $\because f\in C[a,b]$ $\therefore\exists x_0,x_1\in[a,b]$, $f(x_0)\leq f(x)\leq f(x_1)$ $\forall x\in[a,b]$

$$\Rightarrow\int_a^b f(x_0)g(x)\leq\int_a^b f(x)g(x)\leq\int_a^b f(x_1)g(x)\quad\Rightarrow f(x_0)\int_a^b g\leq\int_a^b fg\leq f(x_1)\int_a^b g$$

[化为常数就不参与积分]

介值Thm $\Rightarrow\exists\xi$, $f(\xi)=\frac{\int_a^b fg}{\int_a^b g}$

(2) 第二积分中值 Thm (简单版): $f\in C[a,b]$, g $[a,b]$ 单调且可导, 证:

$\exists\xi\in[a,b]$, s.t. $\int_a^b f(x)g(x)dx=g(a)\int_a^\xi f(x)dx+g(b)\int_\xi^b f(x)dx$

证: 设 $\xi\in[a,b]$, ~~$[a,\xi]$ 上 f $\exists\max=f(x_1)$, $\min=f(x_2)$, $[\xi,b]$ 上 f $\exists\max=f(x_3)$, $\min=f(x_4)$~~

$\forall x\in[a,\xi]$ $g(a)\leq g(x)\leq g(\xi)$ $\Rightarrow g(a)\int_a^\xi f(x)\leq\int_a^\xi f(x)g(x)dx\leq g(\xi)\int_a^\xi f(x)$ f符号未知

$\forall x\in[\xi,b]$ $g(\xi)\leq g(x)\leq g(b)$ $\Rightarrow g(\xi)\int_\xi^b f(x)\leq\int_\xi^b f(x)g(x)\leq g(b)\int_\xi^b f(x)$

$$\Rightarrow g(a)\int_a^\xi f+g(\xi)\int_\xi^b f\leq\int_a^b fg\leq g(\xi)\int_a^\xi f+g(b)\int_\xi^b f$$

$\because g(a)\int_a^\xi f+g(b)\int_\xi^b f\in[g(a)\int_a^\xi f+g(\xi)$

KOKUYO

法二) $\int_a^b fg = \int_a^b (\int_a^x f)' \cdot g = (\int_a^x f) \cdot g \Big|_a^b - \int_a^b \underbrace{\int_a^x f}_{F(x)} \cdot g(x)\,dx$ 设F(x)强行分部 ⇒出现g'

$= g(b) \int_a^b f(x)\,dx - \int_a^b F(x) g'(x)\,dx$ 积分 $= F(\xi) \int_a^b g'(x)\,dx$

$= g(b) \int_a^b f(x)\,dx - F(\xi)(g(b) - g(a))$ g'不变号 ∴可用第一中值

$= g(b) \int_\xi^b f(x)\,dx + g(a) \int_\xi^a f(x)\,dx$

4. (Riemann - Lebesgue引理) 设 $f \in C'[a,b]$, 证:

(1) $\lim\limits_{\lambda\to\infty} \int_a^b f(x)\cos\lambda x\,dx = 0$ $A \in \mathbb{R}$

(2) $\lim\limits_{\lambda\to\infty} \int_a^b f(x)\sin\lambda x\,dx = 0$ $B \in \mathbb{R}$

$A + Bi = 0 \Rightarrow A = B = 0$

(1) $\int_a^b f(x) e^{i\lambda x}\,dx = \int_a^b f(x)(\cos\lambda x + i\sin\lambda x)\,dx$

$\int_a^b f(x) \cdot (e^{i\lambda x})' \cdot \frac{1}{\lambda i} \cdot dx = \underbrace{\frac{-i}{\lambda}}_{\to 0} \cdot \Big(\underbrace{e^{i\lambda x}}_{\in[-\sqrt{2},\sqrt{2}]\ \text{有界}} \cdot f(x) \Big|_a^b - \int_a^b e^{i\lambda x} \cdot f'(x)\,dx \Big)$

改: R-L引理: $f \in R[a,b]$, g 周期=T且$\in R[0,T]$ 则:

$$\lim_{n\to\infty} \int_a^b f(x) g(nx)\,dx = \frac{1}{T} \int_0^T g(x)\,dx \cdot \int_a^b f(x)\,dx$$

只证 $g \geq 0$ 恒成立, a,b 整除T的情形

$$\int_0^T f(x) g(nx)\,dx = \sum_{k=1}^{n} \int_{\frac{k-1}{n}T}^{\frac{k}{n}T} f(x) g(nx)\,dx = \sum_{k=1}^{n} f(\xi_k) \int_{\frac{k-1}{n}T}^{\frac{k}{n}T} g(nx)\,dx$$

$$\therefore \lim_{n\to\infty} \int_0^T f(x) g(nx)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(\xi_k) \int_{\frac{k-1}{n}T}^{\frac{k}{n}T} g(nx)\,dx = \int_0^T g(x)\,dx \cdot \lim_{n\to\infty} \sum_{k=1}^{n} f(\xi_k) \cdot \frac{T}{n} \cdot \frac{1}{T}$$

$$= \frac{1}{T} \int_0^T g \cdot \int_0^T f$$

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5. 证：$\lim\limits_{n\to\infty}\int_0^{\frac{\pi}{2}}\sin^n x\,dx=0$

错1：$\neq\int_0^{\frac{\pi}{2}}\lim\limits_{n\to\infty}\sin^n x\,dx$ 不能换序

错2：$=\lim\limits_{n\to\infty}\sin^n\xi\int_0^{\frac{\pi}{2}}1\cdot dx$

ξ为ξ_n $\therefore\lim\limits_{n\to\infty}\sin^n\xi_n$不一定$=0$

错又正解：$\varepsilon-\delta$语言（定义）

根据积分上下界对放进行放缩

$\forall\varepsilon>0,\ \int_0^{\frac{\pi}{2}}\sin^n x\,dx=\int_0^{\frac{\pi}{2}-\varepsilon}+\int_{\frac{\pi}{2}-\varepsilon}^{0}<\int_0^{\frac{\pi}{2}-\varepsilon}\sin^n(\tfrac{\pi}{2}-\varepsilon)\,dx+\int_{\frac{\pi}{2}-\varepsilon}^{\frac{\pi}{2}}\sin\tfrac{\pi}{2}\,dx$

$=\underline{(\tfrac{\pi}{2}-\varepsilon)\sin^n(\tfrac{\pi}{2}-\varepsilon)+\varepsilon}$ n充分大时，$\simeq\varepsilon<\varepsilon$ ✓

$\therefore\lim\limits_{n\to\infty}\int_0^{\frac{\pi}{2}}\sin^n x\,dx=0$

法二 $\int_0^{\frac{\pi}{2}}\sin^n x\,dx=\int_0^{\frac{\pi}{2}}(\cos x)'\cdot\sin^{n-1}x\,dx=-\cos x\sin^{n-1}x\Big|_0^{\frac{\pi}{2}}+\int_0^{\frac{\pi}{2}}\cos x\cdot\sin^{n-2}x\cdot\cos x\,dx$

$\Rightarrow I_n=\frac{n-1}{n}I_{n-2}\Rightarrow$ $I_{2m}=\frac{(2m-1)!!}{(2m)!!}=\frac{2m-1}{2m}\cdot\frac{2m-3}{2m-2}\cdot\frac{2m-5}{2m-4}\cdots\frac{1}{2}$

$I_{2m-1}=\frac{(2m-2)!!}{(2m-1)!!}$

$(I_{2m})^2<\frac{2m-1}{2m}\cdot\frac{2m}{2m+1}\cdot\frac{2m-3}{2m-2}\cdot\frac{2m-2}{2m-1}\cdots\frac{1}{2}\cdot\frac{2}{3}$

$\frac{2m}{2m+1}\cdot\frac{2m-1}{2m}\cdot\frac{2m-2}{2m-1}\cdot\frac{2m-3}{2m-2}\cdots\frac{2}{3}\cdot\frac{1}{2}=\frac{1}{2m+1}\Rightarrow I_{2m}<\sqrt{\frac{1}{2m+1}}\to0$

I_{2m-1}同理

6.（定义法 Cauchy判敛）

设f在$[0,+\infty)$上的每个闭区间上可积，$a>1$，且$\int_a^{+\infty}xf(x)$收敛，证：$\int_a^{+\infty}f(x)\,dx$收敛

没说$f\geq0$ ∴比较判别法✗

另：$\int_a^b f=\int_a^b\underline{xf(x)}\cdot\frac{1}{x}$（可能变号）

但可以改进：∵$\int_a^{+\infty}xf(x)\,dx$收敛，∴$\forall\varepsilon>0,\exists M>1$，s.t. $\forall B>A>M$，$\left|\int_A^B xf(x)\,dx\right|<\varepsilon$

∵$f\in R[A,B]$，$\forall[A,B]\subseteq[0,+\infty)$ ∴$xf(x)\in R[A,B]$，$\forall[A,B]\subseteq[0,+\infty)$ ~~$f(x)$~~在$[A,B]$所有闭区间上可积

⇒~~可把~~

7. 设 $f,g\in C$，证：(1) $\int_0^{\pi} g(x)f(\sin x)dx=\int_0^{\frac{\pi}{2}}(g(x)+g(\pi-x))\cdot f(\sin x)dx$

证：$\int_0^{\pi} g(x)f(\sin x)dx=\int_0^{\frac{\pi}{2}} g(x)f(\sin x)dx+\int_{\frac{\pi}{2}}^{\pi} g(x)f(\sin x)dx$

第二项，$t=\pi-x$ $\int_0^{\frac{\pi}{2}} g(x)f(\sin x)dx+\int_0^{\frac{\pi}{2}} g(\pi-t)f(\sin t)dt=\int_0^{\frac{\pi}{2}}(g(x)+g(\pi-x))\cdot f(\sin x)dx$

(2) 求 $\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$

法一 $=\int_0^{\pi} x\cdot\frac{\sin x}{2-\sin^2 x}dx=\int_0^{\frac{\pi}{2}}\pi\cdot\frac{\sin x}{2-\sin^2 x}dx=\pi\int_0^{\frac{\pi}{2}}\frac{-d(\cos x)}{1+\cos^2 x}=-\pi\cdot\arctan(\cos x)\Big|_0^{\frac{\pi}{2}}$

$=\frac{\pi^2}{4}$

利用三角函数对称性

法二 $=I=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}$ $x\to\pi-x$ $\int_0^{\pi}\frac{(\pi-x)\sin x}{1+\cos^2 x}dx=\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx-I\Rightarrow I=\frac{\pi^2}{4}$

法三 $\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx=\int_0^{\pi}\frac{(x-\frac{\pi}{2})\sin x}{1+\cos^2 x}dx+\int_0^{\pi}\frac{\frac{\pi}{2}\sin x}{1+\cos^2 x}dx=\frac{\pi}{2}\int_0^{\pi}\frac{\sin x\,dx}{1+\cos^2 x}$

拆出"奇"函数　关于$(\frac{\pi}{2},0)$中心对称 $\therefore\int=0$

8. 设 $f\in C$. 证：(1) $\int_0^{\sqrt[n]{a}} x^{n-1}f(x^n)dx=\frac{1}{n}\int_0^a f(x)dx$

(2) $\int_0^a\left(\int_0^x f\right)dx=\int_0^a f(x)(a-x)dx$　(Fubini)

(3) $\int_0^a\left(\int_0^x f(x)f(y)dy\right)dx=\frac{1}{2}\left(\int_0^a f\right)^2$　(Gauss)

证：(1) 左 $=\int_0^{\sqrt[n]{a}} f(x^n)\cdot d(x^n)\cdot\frac{1}{n}=n\int_0^a f(x)dx$

(2) 设 $F(x)=\int_0^x f$　左 $=\int_0^a F'(x)(a-x)dx=F(x)(a-x)\Big|_0^a+\int_0^a F(x)dx$

$=\int_0^a\left(\int_0^x f\right)dx$

(3) $\int_0^a\left(\int_0^x f(x)f(y)dy\right)dx=\int_0^a f(x)\cdot\left(\int_0^x f\right)\cdot dx=\int_0^a F'(x)F(x)dx=\frac{1}{2}F^2(x)\Big|_0^a$

对 dy 来说相当于系数 $\therefore$ 可拿出去　$=\frac{1}{2}\left(\int_0^a f\right)^2$

9. 设 $f[a,b]$ 上可导且 $f'(a)=L>0$，由积分中值 Thm 知，$\forall a<x\le b$，$\exists \xi(x)\in[a,x]$ s.t.

$\int_a^x f = f(\xi(x))\cdot(x-a)$　(1) 证：$\forall \varepsilon\in(0,L)$，$\exists \delta>0$，s.t. $\forall a<x<a+\delta$，有

$\frac{L-\varepsilon}{2(L+\varepsilon)} < \frac{\xi(x)-a}{x-a} < \frac{L+\varepsilon}{2(L-\varepsilon)}$　(2) 计算 $\lim\limits_{x\to a^+}\frac{\xi(x)-a}{x-a}$

(1) 取 $0<\varepsilon'<\varepsilon$，$\because f'(a)=L$ $\therefore \lim\limits_{t\to a^+}\frac{f(t)-f(a)}{x-a}=L$ 对 ε'，$\exists\delta>0$，$\forall a<t<a+\delta$.

$L-\varepsilon' < \frac{f(t)-f(a)}{t-a} < L+\varepsilon' \Rightarrow f(a)+(L-\varepsilon')(t-a)\le f(t)\le f(a)+(L+\varepsilon')(t-a)$ ①

$\Rightarrow \forall x\in(a,a+\delta)$，$\int_a^x[f(a)+(L-\varepsilon')(t-a)]\,dt \le \int_a^x f \le \int_a^x[f(a)+(L+\varepsilon')(t-a)]\,dt$ ②

$\because \xi(x)\in[a,x]\subseteq[a,a+\delta)$ $\therefore$ 可将 ① 式中 t 代成 $\xi(x)$.

$\Rightarrow f(a)+(L-\varepsilon')(\xi(x)-a)\le f(\xi(x))\le f(a)+(L+\varepsilon')(\xi(x)-a)$ (由①)

$f(a)+(L-\varepsilon')\frac{x-a}{2} \le \frac{1}{x-a}\int_a^x f \le f(a)+(L+\varepsilon')\frac{x-a}{2}$ (由②)

$\Rightarrow \frac{L-\varepsilon}{2(L+\varepsilon)} < \frac{L-\varepsilon'}{2(L+\varepsilon')} \le \frac{\xi(x)-a}{x-a} \le \frac{L+\varepsilon'}{2(L-\varepsilon')} < \frac{L+\varepsilon}{2(L-\varepsilon)}$ ✓

(2) 由(1)，$\frac{L-\varepsilon}{2(L+\varepsilon)} < \frac{\xi(x)-a}{x-a} < \frac{L+\varepsilon}{2(L-\varepsilon)}$

$\because \lim\limits_{\varepsilon\to0}\frac{L-\varepsilon}{2(L+\varepsilon)} = \lim\limits_{\varepsilon\to0}\frac{L+\varepsilon}{2(L-\varepsilon)} = \frac{1}{2}$ $\because$ 对上述 ε，$\exists\delta>0$ 对 $\forall a<x<a+\delta$.

$\frac{1}{2}-\eta < \frac{L-\varepsilon}{2(L+\varepsilon)} < \frac{L+\varepsilon}{2(L-\varepsilon)} < \frac{1}{2}+\eta$

有 $\frac{L-\varepsilon}{2(L+\varepsilon)} < \frac{\xi(x)-a}{x-a} < \frac{L+\varepsilon}{2(L-\varepsilon)}$ $\therefore \lim\limits_{x\to a^+}\frac{\xi(x)-a}{x-a}=\frac{1}{2}$

10. (1) $\int_0^{\frac{\pi}{4}}\frac{1}{1-\tan x}dx$ 敛散性

(2) 计算 $\int_0^{+\infty}\frac{x-x^2+x^3-x^4+\cdots-x^{2020}}{(1+x)^{2023}}dx$

(1) $\int_0^{\frac{\pi}{4}}\frac{1}{1-\tan x}dx = \int_0^{\frac{\pi}{4}}\frac{\cos^2x+\sin x\cos x}{\cos^2x-\sin^2x} = \int_0^{\frac{\pi}{4}}\frac{\cos2x+1+\sin2x}{2\cos2x}dx = \frac{1}{2}\int_0^{\frac{\pi}{4}}\left(\tan2x+\frac{1}{\cos2x}\right)$

不那么显然，除非积出来/比较判别 $> \tan2x$

$\because \frac{1}{2}\int_0^{\frac{\pi}{4}}\tan2x\,dx$ 发散 $\therefore \int_0^{\frac{\pi}{4}}\frac{1}{1-\tan x}dx$ 发散

法1 $\lim\limits_{x\to\frac{\pi}{4}^-}\frac{\frac{1}{\frac{\pi}{4}-x}}{\frac{1}{1-\tan x}} = \lim\limits_{x\to\frac{\pi}{4}^-}\frac{1-\tan x}{\frac{\pi}{4}-x} \overset{\frac{0}{0}}{=\!=} 2$ $\therefore \int_0^{\frac{\pi}{4}}\frac{dx}{1-\tan x}$ 与 $\int_0^{\frac{\pi}{4}}\frac{dx}{\frac{\pi}{4}-x}$ 同敛散 ✓

发散KOKUYO

(2) $\int_0^{+\infty} \frac{x-x^2+\cdots-x^{2024}}{(1+x)^{2025}}dx = \sum_{k=1}^{2024}(-1)^{k-1}\cdot\int_0^{+\infty}\frac{x^k}{(1+x)^{2025}}dx$

$\because \lim_{x\to+\infty}\frac{\frac{x^k}{(1+x)^{2025}}}{x^{k-2025}}=1$ 且 $\int_0^{+\infty}x^{k-2025}$ 收敛 $(1\le k\le 2024)$ $\therefore$ 原收敛

$\int_0^{+\infty}\frac{x^k}{(1+x)^{2025}}dx = \lim_{a\to0^+}\int_a^1\frac{x^k}{(1+x)^{2025}}dx + \lim_{b\to+\infty}\int_1^b\frac{x^k}{(1+x)^{2025}}dx$ 换元 $x=t$. 找特点.

$-\left(\lim_{a\to0^+}\int_t^1\frac{t^{2024-k}}{(1+t)^{2025}}dt+\lim_{b\to+\infty}\int_1^t\frac{t^{2024-k}}{(1+t)^{2025}}dt\right)$ $\because$ 都收敛 $-\left(\int_{+\infty}^1+\int_1^0\right)=\int_0^{+\infty}\frac{t^{2024-k}}{(1+t)^{2025}}d$

$\Rightarrow\int_0^{+\infty}\frac{x^k-x^{2024-k}}{(1+x)}dx=0$ $\therefore$ 原 $=0$

11. $f\in C[a,b]$. $f(a)=0$ (1) 证 $\int_a^b f^2\le\frac{(b-a)^2}{2}\int_a^b(f')^2$

(2) if $f(b)=0$, $\int_a^b f^2\le\frac{(b-a)^2}{8}\int_a^b(f')^2$

证: (1) $\int_a^x f'=f(x)-f(a)=f(x)$

$\Rightarrow f^2=(\int_a^x f'\cdot 1)^2\le\int_a^x(f')^2\cdot\int_a^x 1^2\le\int_a^b(f')^2\cdot(x-a)$

(不一定 x) $\le\int_a^b(f')^2$

$\Rightarrow\int_a^b f^2\le\int_a^b(f')^2\cdot\int_a^b(x-a)=\frac{(b-a)^2}{2}\int_a^b(f')^2$

(2) $\int_a^x f'=f(x)\Rightarrow f^2=(\int_a^x f'\cdot1)^2\le\int_a^x(f')^2(x-a)\Rightarrow\int_a^{\frac{a+b}{2}}f^2\le\int_a^{\frac{a+b}{2}}(x-a)dx\cdot(\int_a^{\frac{a+b}{2}}(f$

$\int_x^b f'=-f(x)\Rightarrow\int_{\frac{a+b}{2}}^b f^2\le\frac{(b-a)^2}{8}\int_{\frac{a+b}{2}}^b(f')^2$ $=\frac{(b-a)^2}{8}\int_a^{\frac{a+b}{2}}(f')^2$

$\Rightarrow\int_a^b f^2\le\frac{(b-a)^2}{8}\int_a^b(f')^2$

Date 2024 : 1. 11

周期函数积分

1. 设 $f\in C[0,1]$, $g\geq 0$ 且 $T=1$, $g\in R[0,1]$, 证:

$$\lim_{n\to\infty}\int_0^1 f(x)g(nx)\,dx=\left(\int_0^1 f(x)\,dx\right)\cdot\left(\int_0^1 g(x)\,dx\right)$$

积分中值

证: $\int_0^1 f(x)g(nx)\,dx=\sum_{k=1}^{n}\int_{\frac{k-1}{n}}^{\frac{k}{n}} f(x)g(nx)\,dx=\sum_{k=1}^{n} f(\xi_k)\int_{\frac{k-1}{n}}^{\frac{k}{n}} g(nx)\,dx$ $\quad \xi_k\in\left[\frac{k-1}{n},\frac{k}{n}\right]$

$\lim$, 令 $nx=t$ $\quad dx=\frac{1}{n}dt$ $\quad \lim\sum_{k=1}^{n} f(\xi_k)\cdot\frac{1}{n}\cdot\int_0^1 g=\left(\int_0^1 f\right)\cdot\left(\int_0^1 g\right)$

无限细分 $\quad$ 选点 $\quad \Delta x_i$

2. 设 $f\in C[0,2\pi]$, 证: $\lim_{n\to\infty}\int_0^{2\pi} f(x)|\sin nx|\,dx=\frac{2}{\pi}\int_0^{2\pi} f(x)\,dx$

证: $\lim_{n\to\infty}\int_0^{2\pi} f(x)|\sin nx|\,dx=\lim_{n\to\infty}\sum_{k=1}^{n} f(\xi_k)\cdot\int_{\frac{2k-2}{n}\pi}^{\frac{2k}{n}\pi}|\sin nx|\,dx=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{4}{n} f(\xi_k)$

$=\lim_{n\to\infty}\frac{2}{\pi}\sum_{k=1}^{n}\frac{2\pi}{n} f(\xi_k)=\frac{2}{\pi}\int_0^{2\pi} f$ $\quad \xi_k\in\left[\frac{2k-1}{n}\pi,\frac{2k\pi}{n}\right]$ $\quad \Delta x_i=\frac{2\pi}{n}$

No.

Date

广义积分敛散性

1. 设 $f\in C[0,+\infty)$ $g(x)=\int_0^x f$

(1) 证: $\int_A^B \frac{g^2(x)}{x^2}dx = \frac{g^2(A)}{A} - \frac{g^2(B)}{B} + 2\int_A^B \frac{f(x)g(x)}{x}dx \quad (A,B\in(0,+\infty))$

(2) 计 $\int_0^{+\infty} f^2$ 收, 证: $\int_0^{+\infty}\frac{g^2(x)}{x^2}$ 收且 $\int_0^{+\infty}\frac{g^2(x)}{x^2}dx \le 4\int_0^{+\infty} f^2$

(2) 证: $\because \lim_{x\to 0^+}\frac{g^2(x)}{x} \overset{\frac{0}{0}}{=\!=} \lim_{x\to 0^+} 2g(x)\cdot f(x) = 0\cdot 2f(0) = 0$

$\therefore F(B)=\int_0^B \frac{g^2(x)}{x^2} \overset{用(1)}{=\!=} -\frac{g^2(B)}{B} + 2\int_0^B \frac{f(x)g(x)}{x}dx$

先证收敛: $\because F(B)\uparrow$ $\therefore$ 即证 $F(B)$ 有上界

$$F(B) = -\frac{g^2(B)}{B} + 2\underbrace{\int_0^B \frac{f(x)g(x)}{x}dx}_{\text{Cauchy}} \le 0 + 2\left(\int_0^B f^2\right)^{\frac{1}{2}}\cdot\underbrace{\left(\int_0^B \frac{g^2}{x^2}\right)^{\frac{1}{2}}}_{=F(B)}$$

$\Rightarrow F(B)\le 4\int_0^B f^2 \Rightarrow F(B)$ 收

$\Rightarrow \int_0^{+\infty}\frac{g^2}{x^2}\le 4\int_0^{+\infty} f^2$

引理1 (Dirichlet test)：设 $F(t)=\int_a^t f$ $[a,+\infty)$ 有界，$g(x)$ $(a,+\infty)$ 单调且 $\lim_{x\to+\infty} g(x)=0$，

则 $\int_a^{+\infty} f(x)g(x)dx$ 收敛

证：$\forall A_2>A_1\geq a$，由第二积分中值 Thm，$\exists \xi\in[A_1,A_2]$，

$$\int_{A_1}^{A_2} fg = g(A_1)\int_{A_1}^{\xi} f + g(A_2)\int_{\xi}^{A_2} f$$

设 $|F(t)|\leq C$，则 $\left|\int_{A_1}^{\xi} f\right| = |F(\xi)-F(A_1)| \leq 2C$ 同理，$\left|\int_{\xi}^{A_2} f\right| \leq 2C$

$\because \lim_{x\to+\infty} g(x)=0 \therefore \forall \varepsilon$，$\exists M$，$\forall A_2>A_1>M$，$|g(A_1)|<\frac{\varepsilon}{4C}$，$|g(A_2)|<\frac{\varepsilon}{4C}$

$\Rightarrow \forall \varepsilon>0$，$\exists M>a$，$\forall A_2>A_1>M$，$\left|\int_{A_1}^{A_2} fg\right|<\varepsilon$ ✓

引理2 (Abel test) 设 $\int_a^{+\infty} f$ 收敛，$g(x)$ 在 $[a,+\infty)$ 单调有界，则 $\int_a^{+\infty} fg$ 收敛

KOKUY